

YUE CAO

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Department of Mechanical Engineering ♦ University of Michigan, Ann Arbor, MI 48109

RESEARCH FOCUS

My research advances autonomous and intelligent manufacturing systems by integrating generative artificial intelligence with process monitoring, control, and human-robot collaboration. I develop physics-informed and data-efficient manufacturing systems designed for robust and safe operation in welding, additive manufacturing, and automotive stamping.

Key words: Intelligent manufacturing, arc welding, wire arc additive manufacturing, generative AI, physics-informed machine learning, process monitoring and control, human–robot collaboration, robot learning from demonstration

ACADEMIC APPOINTMENTS

Postdoctoral Research Fellow

Apr 2026 – Present

Department of Mechanical Engineering, University of Michigan, Ann Arbor, MI

Advisor: Prof. Chenhui Shao

EDUCATION

Ph.D. in Electrical Engineering

Jan 2023 – May 2026

University of Kentucky, Lexington, KY

Dissertation: *Robotizing Complex Welding Processes Through Imitation Learning and Generative Models from Human Demonstration*

Advisor: Prof. Yuming Zhang

M.Eng. in Materials Processing Engineering

Sep 2018 – Jun 2021

Tianjin University, Tianjin, China

Advisor: Prof. Zhijiang Wang

B.Eng. in Materials Forming and Control Engineering

Sep 2014 – Jun 2018

Tianjin University, Tianjin, China

HONORS AND AWARDS

Henry Granjon Prize, International Institute of Welding (IIW)

2025

Category D: Human Related Subjects

- Awarded for the paper “Control of DE-GMAW through Human–Robot Collaboration” [J7]; presented at the 78th IIW Annual Assembly in Genoa, Italy.
- IIW’s annual international prize, sponsored by its French Delegation, for research papers by young welding researchers; one winner is named in each of four categories. Official listing: [2025 IIW Awards booklet](#).

A. F. Davis Silver Medal, American Welding Society (AWS)

2024

Category I: Design of Welding Processes, Machines, and Equipment

- Awarded for the paper “Effects of Filler Wire Intervention on the Gas Tungsten Arc: Part III—Process Stability Control of Wire-Filled GTAW” [J22], coauthored with S. Zou, Z. Wang, and S. Hu.
- AWS national award given each year to the authors of up to three Welding Journal papers judged the best contributions to the progress of welding. Official listing: [AWS National Awards recipients](#).

PROPOSAL WRITING

NSF CMMI Collaborative Research

2026 – 2029

Unified Sensing and Control for Generalizable Real-Time Arc Welding

PI: Prof. Yuming Zhang (University of Kentucky); Co-PI: Prof. Hai Lin (University of Notre Dame)

Role: Proposal conceptualization and primary technical writing

Result: Funded by NSF CMMI, \$325,000

RESEARCH EXPERIENCE

University of Michigan

Apr 2026 – Present

Stamping xAI: Trustworthy Process Intelligence for Automotive Stamping (General Motors collaboration)

- Developing an interpretable representation of press-tonnage signals for unsupervised detection of rare stamping defects.
- Extending it to Bayesian test-time adaptation for reliable monitoring under process drift.

Molten Pool Dynamics Modeling and Autonomous Robotic Wire Arc Additive Manufacturing

- Developing a physics-structured neural state-space model that captures molten pool dynamics across GTAW, GMAW, and WAAM.
- Leading the design of a multisensor robotic WAAM platform for closed-loop control of the melt pool and part geometry.

University of Kentucky

Jan 2023 – Mar 2026

Generative Modeling and Intelligent Monitoring of Arc Manufacturing

- Developed generative models that learn a compact latent state of the weld pool and predict its evolution under process inputs.
- Established a Bayesian online monitoring framework that estimates hidden weld states and transfers across arc welding processes.
- Proposed self-correcting label refinement for reliable anomaly detection in wire arc additive manufacturing.
- Reconstructed molten pool flow fields from high-speed video with physics-informed neural networks.

Human–Robot Collaboration and Skill Transfer for Welding Robots

- Built a VR-based dual-robot collaboration system through which welders remotely control complex double-electrode GMAW.
- Added generative weld pool prediction to the interface so operators can act before deviations occur.
- Learned welders' control strategies from demonstrations and transferred them to autonomous robotic welding.

Tianjin University

Sep 2018 – Jun 2021

Arc-Signal Sensing and Closed-Loop Control of Weld Penetration

- Identified arc-voltage features that reveal weld penetration, enabling low-cost sensing from built-in electrical signals.
- Developed a reduced-order nonlinear model and adaptive model predictive control of backside weld width.

PUBLICATIONS

Google Scholar: 465 citations, h-index 13 (as of September 23, 2026)

Journal Articles: First Author

- [J1] **Cao, Y.**, Ma, N., Ye, Q., & Zhang, Y. Human predictive control of welding process through generative AI enhanced human–robot collaboration. *Journal of Manufacturing Processes*. Accepted. [Code on GitHub](#).
- [J2] **Cao, Y.**, Su, Q., Hoang Tien, D., & Nguyen, V. A. Reconstruction of molten pool flow fields from high-speed video using physics-informed neural networks. *Journal of Intelligent Manufacturing*. Under review (submitted April 2026). [Code on GitHub](#).

- [J3] **Cao, Y.**, Wang, Z., Hu, S., & Wang, T. (2023). Adaptive predictive control of backside weld width in pulsed gas metal arc welding using electrical characteristic signals as feedback. *IEEE Transactions on Control Systems Technology*, 31(6), 2879–2886.
- [J4] **Cao, Y.**, Lin, H., & Zhang, Y. (2026). Robust monitoring of arc welding processes: A generalizable framework with DVAE and particle filter. *Journal of Manufacturing Processes*, 160, 15–27.
- [J5] **Cao, Y.**, Mattera, G., Wang, P., & Zhang, Y. (2026). Enhancing in-situ anomaly detection during wire arc additive manufacturing process through iterative automated label refinement. *Journal of Manufacturing Processes*, 167, 119–130.
- [J6] **Cao, Y.**, Ye, Q., & Zhang, Y. (2025). Synthesizing weld pool dynamics via VAE-GAN to enhance human control performance. *Journal of Manufacturing Processes*, 155, 98–107.
- [J7] **Cao, Y.**, & Zhang, Y. (2025). Control of DE-GMAW through human–robot collaboration. *Welding in the World*, 69(5), 1459–1468. **2025 IIW Henry Granjon Prize.**
- [J8] **Cao, Y.**, Zhou, Q., Yuan, W., Ye, Q., Popa, D., & Zhang, Y. (2024). Human–robot collaborative assembly and welding: A review and analysis of the state of the art. *Journal of Manufacturing Processes*, 131, 1388–1403.
- [J9] **Cao, Y.**, Guo, S., & Zhang, Y. (2026). Robotizing GTAW through learning human response. *Welding in the World*, 70, 1361–1371.
- [J10] **Cao, Y.**, Chen, H., & Zhang, Y. (2025). Monitoring of DE-GMAW process in human–robot collaboration. *Welding in the World*, 69(5), 1427–1435.
- [J11] **Cao, Y.**, Ye, Q., & Zhang, Y. (2026). Application of generative adversarial networks (GANs) in intelligent welding manufacturing. *Welding in the World*, 70(4), 1283–1298.
- [J12] **Cao, Y.**, Wang, Z., Hu, S., & Wang, W. (2021). Modeling of weld penetration control system in GMAW-P using NARMAX methods. *Journal of Manufacturing Processes*, 65, 512–524.

Journal Articles: Coauthor

- [J13] Le, B. T. G., Le, Q. H., **Cao, Y.**, DebRoy, T., Oliveira, J., Hoang Tien, D., et al. (2026). Wire arc additive manufacturing: Fundamental, challenges, solutions, and emerging frontiers. *Journal of Manufacturing Processes*, 172, 962–1033.
- [J14] Mattera, G., **Cao, Y.**, Zhang, Y., & Nele, L. (2026). Vision-based online quality tracking in wire arc additive manufacturing via hybrid unsupervised deep learning–statistical process monitoring. *Welding in the World*, 70(4), 1261–1281.
- [J15] Li, T., **Cao, Y.**, Ye, Q., & Zhang, Y. (2025). Generative adversarial networks (GAN) model for dynamically adjusted weld pool image toward human-based model predictive control (MPC). *Journal of Manufacturing Processes*, 141, 210–221.
- [J16] Li, T., **Cao, Y.**, & Zhang, Y. (2024). Analysis of weld pool region constituents in GMAW for dynamic reconstruction through characteristic enhancement and LSTM U-Net networks. *Journal of Manufacturing Processes*, 127, 573–588.
- [J17] Muclari, E., **Cao, Y.**, Ye, Q., & Zhang, Y. (2024). Modeling imaged welding process dynamic behaviors using generative adversarial network (GAN) for a new foundation to monitor weld penetration using deep learning. *Journal of Manufacturing Processes*, 124, 187–195.
- [J18] Yu, R., **Cao, Y.**, Martin, J., Chiang, O., & Zhang, Y. (2024). Deep-learning based supervisory monitoring of robotized DE-GMAW process through learning from human welders. *Welding in the World*, 68(4), 781–791.
- [J19] Yu, R., **Cao, Y.**, Martin, J., Chiang, O., & Zhang, Y. (2024). Robotizing double-electrode GMAW process through learning from human welders. *Journal of Manufacturing Processes*, 109, 140–150.

- [J20] Yu, R., **Cao, Y.**, Chen, H., Ye, Q., & Zhang, Y. (2023). Deep learning based real-time and in-situ monitoring of weld penetration: Where we are and what are needed revolutionary solutions? *Journal of Manufacturing Processes*, 93, 15–46.
- [J21] Mucllari, E., Yu, R., **Cao, Y.**, Ye, Q., & Zhang, Y. (2023). Do we need a new foundation to use deep learning to monitor weld penetration? *IEEE Robotics and Automation Letters*, 8(6), 3669–3676.
- [J22] Zou, S., Wang, Z., **Cao, Y.**, & Hu, S. (2023). Effects of filler wire intervention on the gas tungsten arc: Part III —Process stability control of wire-filled GTAW. *Welding Journal*, 102(3), 53s–68s. **2024 AWS A. F. Davis Silver Medal.**
- [J23] Wang, T., Wang, Z., **Cao, Y.**, Wang, Y., & Hu, S. (2021). A multi-BRIEF-descriptor stereo matching algorithm for binocular visual sensing of fillet welds with indistinct features. *Journal of Manufacturing Processes*, 66, 636–650.
- [J24] Zou, S., Wang, Z., Hu, S., Wang, W., & **Cao, Y.** (2020). Control of weld penetration depth using relative fluctuation coefficient as feedback. *Journal of Intelligent Manufacturing*, 31(5), 1203–1213.
- [J25] Wang, W., Wang, Z., Hu, S., Bai, P., Lu, T., & **Cao, Y.** (2018). Weld pool surface fluctuations sensing in pulsed GMAW. *Welding Journal*, 97(12), 327s–337s.

CONFERENCE PRESENTATIONS

Oral presentations, AWS Professional Program, FABTECH

- [P1] **Cao, Y.**, Mattera, G., & Zhang, Y. Addressing label inaccuracy in WAAM anomaly detection via iterative label refinement and unsupervised feature learning. FABTECH 2025, Chicago, IL.
- [P2] **Cao, Y.**, & Zhang, Y. Unsupervised weld penetration prediction for visual monitoring via guided variational autoencoder. FABTECH 2025, Chicago, IL.
- [P3] **Cao, Y.**, & Zhang, Y. Learning human knowledge for robotizing double-electrode GMAW via generative modeling and imitation learning. FABTECH 2025, Chicago, IL.
- [P4] **Cao, Y.**, Mucllari, E., Zhang, Y., & Ye, Q. Weld penetration prediction using GAN with inaccurate labels. FABTECH 2024, Orlando, FL.
- [P5] **Cao, Y.**, & Zhang, Y. Human–robot collaboration for DE-GMAW. FABTECH 2024, Orlando, FL.
- [P6] **Cao, Y.**, Yu, R., Mucllari, E., Ye, Q., & Zhang, Y. Robotizing double-electrode GMAW by learning from human welders. FABTECH 2023, Chicago, IL.

TEACHING EXPERIENCE

Teaching Assistant , ME 499/599 Data Science for Manufacturing Quality Control <i>University of Michigan</i> ; Instructor: Prof. Chenhui Shao	Fall 2026
Teaching Assistant , EE 415G Electromechanics <i>University of Kentucky</i> ; Instructor: Prof. Yuan Liao	Spring 2025
Teaching Assistant , EE 599 Applied Model Predictive Control <i>University of Kentucky</i> ; Instructor: Prof. Yuming Zhang	Spring 2025

STUDENT MENTORING

UROP Research Mentor, Undergraduate Research Opportunity Program, University of Michigan 2026 – Present
Mentoring undergraduate Marcus Scott on building a digital twin for a robotic metal 3D printer.

Tianpu Li, B.S. Electrical Engineering, University of Kentucky 2024
Supervised research on deep learning for weld pool monitoring; two coauthored journal papers.

Sam Kotter, high school student, Henry Clay High School Fall 2025
Guided a project on deep learning for manufacturing applications.

PROFESSIONAL SERVICE

Principal Reviewer, *Welding in the World*

Reviewer for 60+ manuscripts: *Mechanical Systems and Signal Processing*; *Journal of Manufacturing Processes*; *Engineering Applications of Artificial Intelligence*; *Welding in the World*

INDUSTRY EXPERIENCE

Autonomous Driving Engineer, Calmcar Co., Ltd., Tianjin, China Jul 2022 – Dec 2022
Developed lateral and speed control for autonomous electric vehicles.

Motor Control Engineer, Weichai Power Co., Ltd., Weifang, China Aug 2021 – Jul 2022
Developed PMSM current, torque, and speed control with system identification and adaptive control; designed a disturbance-observer-based hill-hold function validated in vehicle tests.

TECHNICAL EXPERTISE

Intelligent process monitoring: Multi-sensor sensing (vision, electrical signals) with generative and physics-informed models for weld-pool state estimation and anomaly detection.

Process modeling and control: System identification and model predictive/adaptive control for arc welding and wire arc additive manufacturing (WAAM) processes.

Robotic manufacturing systems: Robotic arc welding and WAAM cells; dual-robot coordination; VR teleoperation; and learning from human demonstration (Python, ROS, Unity).

Automotive control systems: PMSM drive control and autonomous-vehicle lateral and speed control; model-based design in MATLAB/Simulink; embedded C/C++; SIL/HIL validation; and in-vehicle testing.